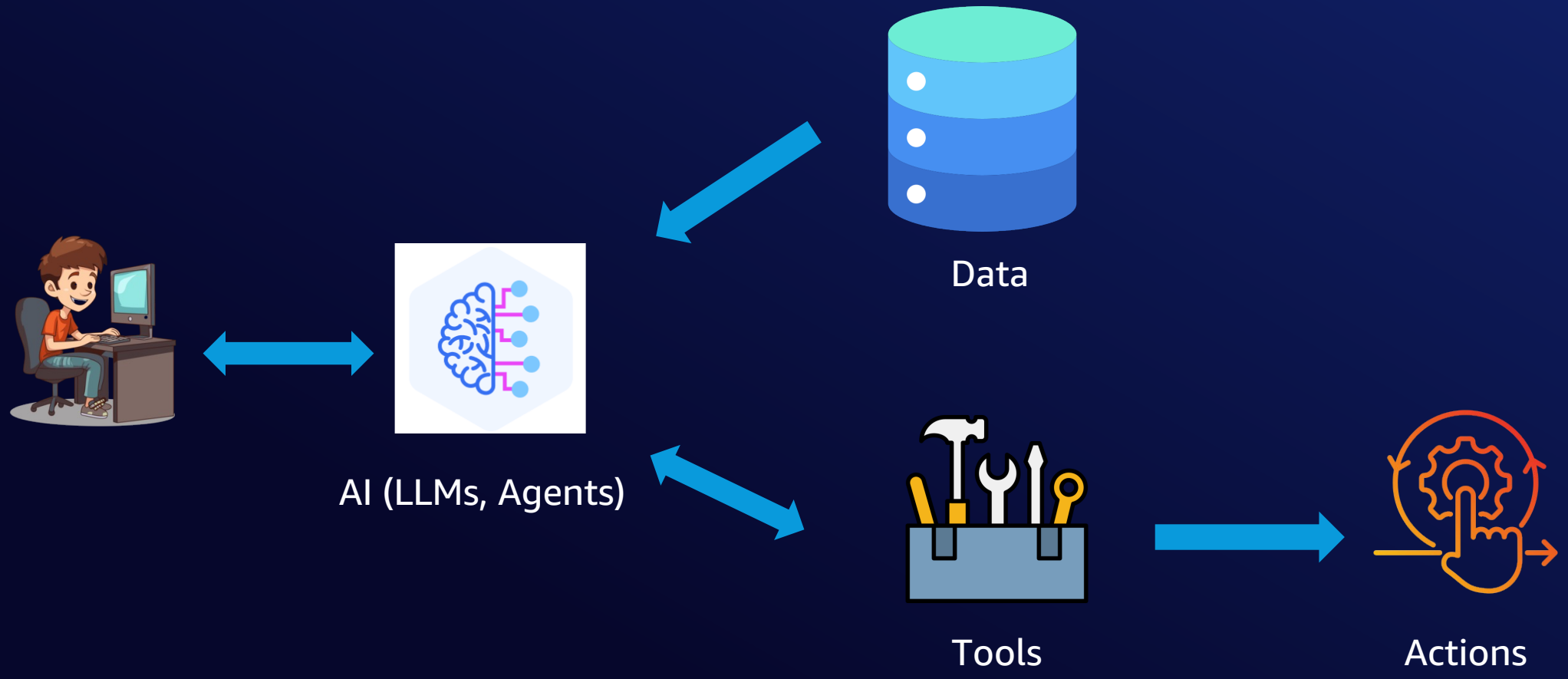
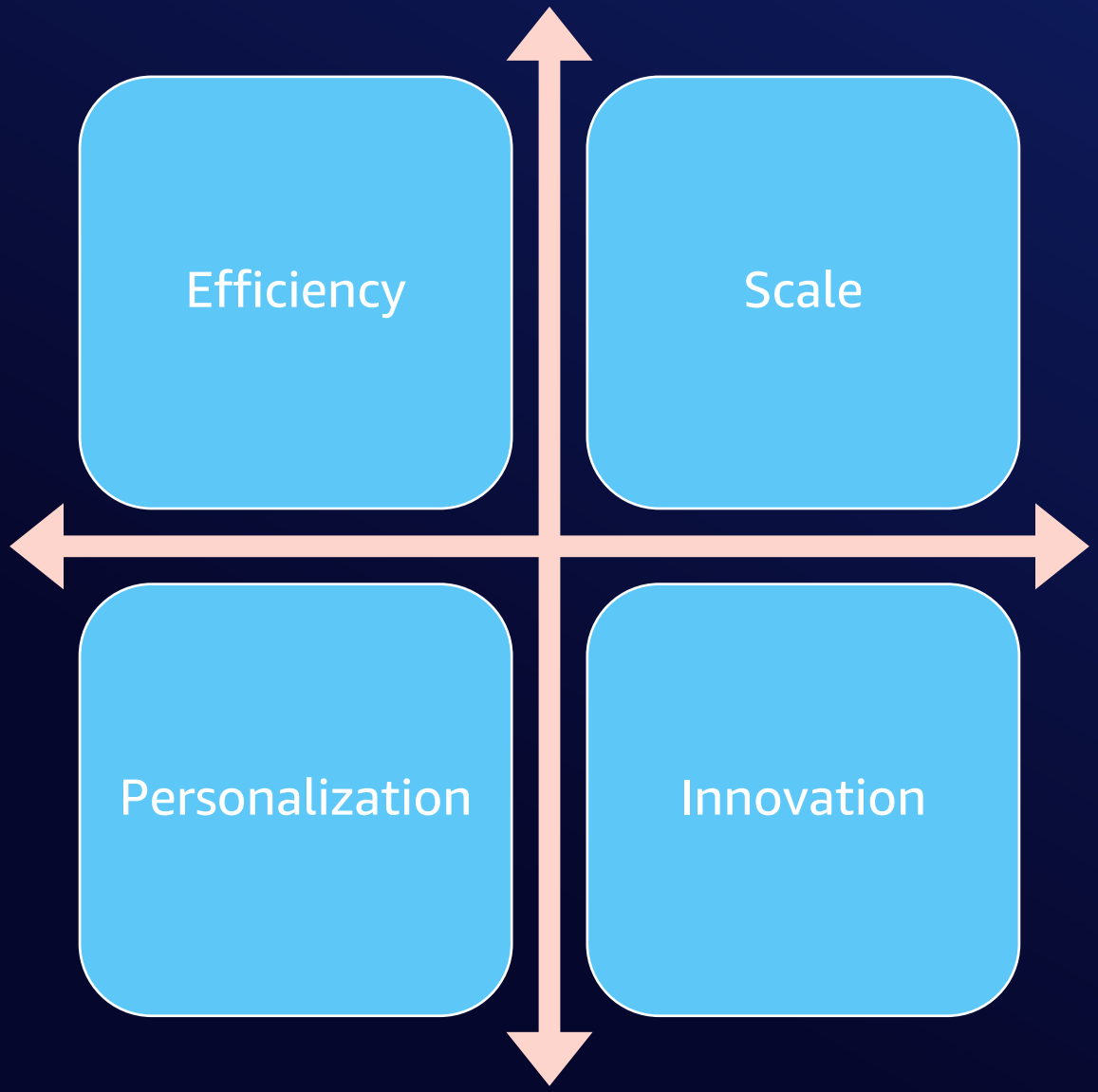
Industry adoption of GenAI and Agentic AI

The “API Moment” of Artificial Intelligence



Why Enterprises Care?



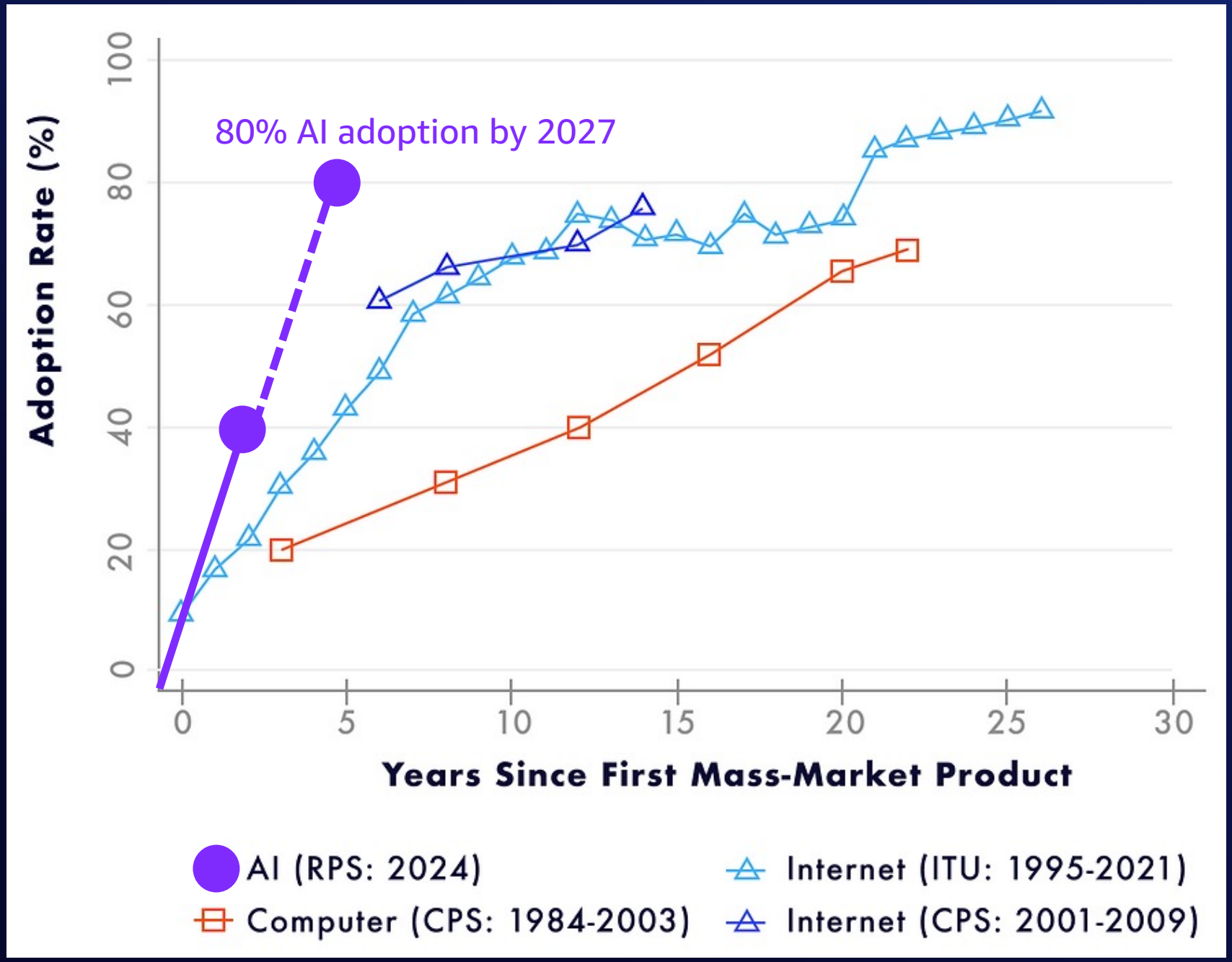
The Executive Conversation Shift

- 2022: “Should we experiment with GenAI?”
- 2023: “How do we implement GenAI safely?”
- 2024: “We need AI agents running our operations”
- 2025: “Our competitors have autonomous workflows—we’re behind”

In less than 2 years, 40% of the US population is using generative AI daily

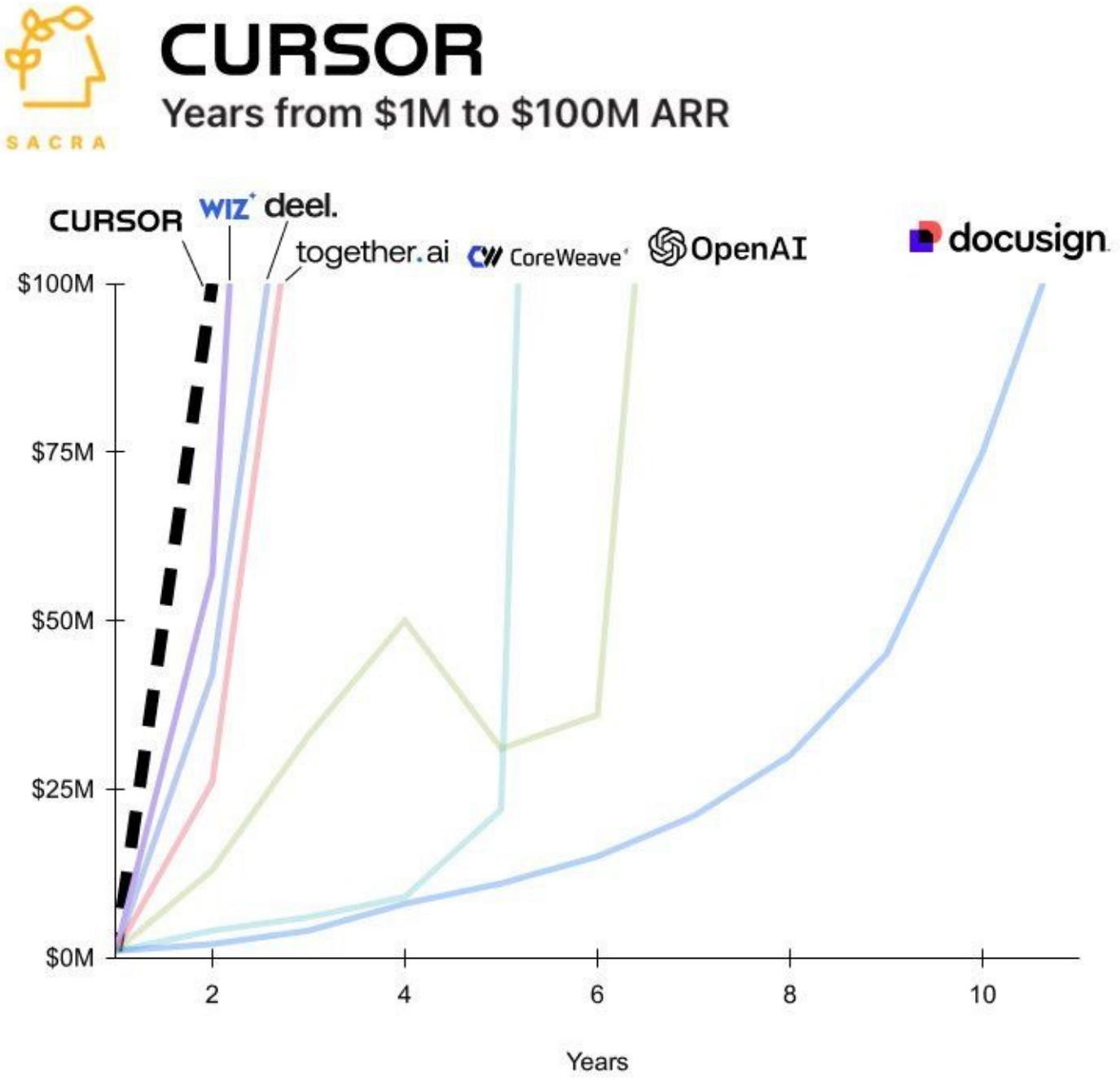
2x faster than the Internet to reach 40%

40% PC adoption took 12 years



9,900%
growth in a single year, smashing the $100M ARR milestone

..then $300M in 4mo and $500M in the next 2mo..



The Success Stories With GenAI and Agentic AI

Financial Services	Technology	M&E	HCLS
• Investment Research • Financial Crime • Customer Services	• Product Design & Engineering • Security • Operations	• Personalization • New Content Development	• Drug Discovery • Medical Diagnosis

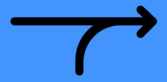
Observations



#1: Technology is Only a Part of the Challenge



#2: Data is Your Competitive Moat

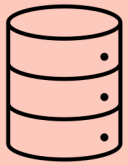


#3: Start Simple, Scale Smart

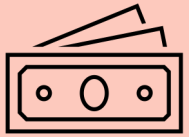


#4: Regulation and Ethics Aren’t Afterthoughts

The Hard Problems to Solve



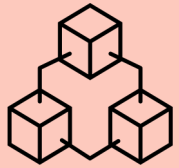
Data Gravity & Governance



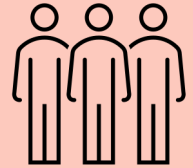
Cost vs. Value



The "Evaluation" Problem



Interoperability and Multi-Agent Systems



Talent & Culture Shift

From Foundation Models to Agentic Mesh

AI Agents are gaining a lot of attention



Video · Jun 10, 2025

Announcing Cisco AI Canvas. Revolutionizing IT with AgenticOps

Autodesk's AI-Driven Transformation: A Strategic Masterstroke in the Design Software Revolution

Eli Grant • Sunday, Aug 31, 2025 7:37 am ET 3min read

Airbnb charts AI-first path as it expands customer service agent

"I think you can't do travel planning without AI going forward," CEO Brian Chesky said.

Published Aug. 8, 2025

Expedia's AI-Driven Surge: A Blueprint for the Future of Travel Tech?

Theodore Quinn • Friday, Aug 8, 2025 9:48 am ET

HPE accelerates self-driving network operations with new Mist agentic AI-native innovations

August 26, 2025

Press Release Software AI Networking

Infor Announces Strategic Collaboration with Amazon Web Services to Propel Generative AI Innovations

Intuit Introduces Ground-Breaking Virtual Team of AI Agents to Fuel Growth for Businesses

July 01, 2025 8:30am EDT

LexisNexis Introduces Protégé General AI and Expands Agentic AI Leadership, Bringing Secure, Integrated Access to General-Purpose AI for Legal Professionals

August 11, 2025

Meta is targeting 'hundreds of millions' of businesses in agentic AI deployment

PUBLISHED THU, MAR 6 2025•11:58 AM EST

Workday Unveils Next Generation of Illuminate Agents to Transform HR and Finance Operations

PINTEREST POSITIONS ITSELF AS AI-ENABLED SHOPPING ASSISTANT AMID ONGOING "CAMBRIAN MOMENT" FOR AI

Nvidia debuts next-gen agentic AI and reasoning robotic models at SIGGRAPH 2025



BY KYT DOTSON

The Agent–Driven Enterprise: Deconstructing 10 AI Use Cases Powering Netflix

Introduction: The Rise of the Specialized Agent

Adobe Reveals New Agentic AI System Of Helpers And Aids For Its Software

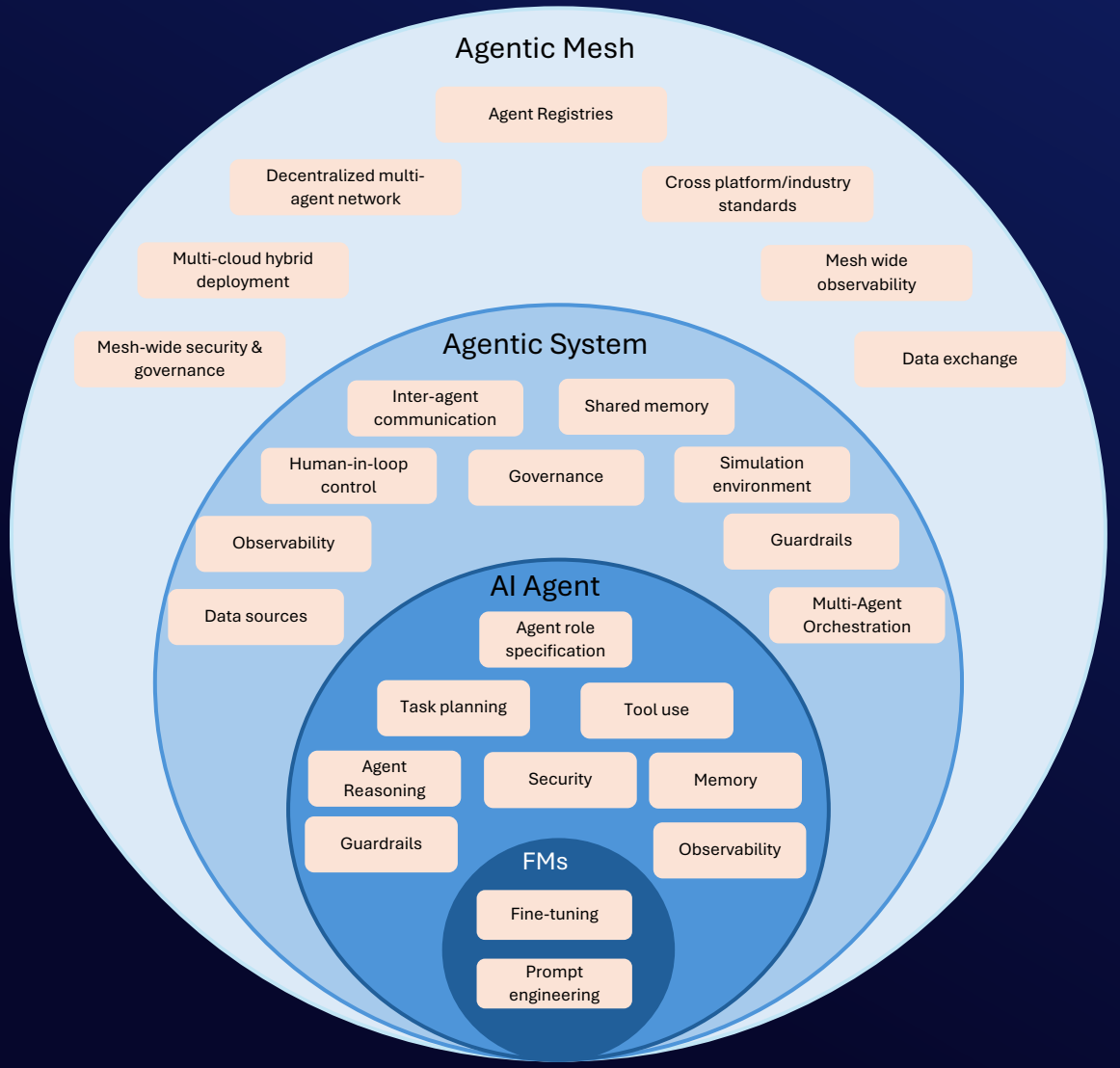
By Mark Sparrow, Senior Contributor. Technolo... Follow Author

Qualtrics AI Powered Innovations Deliver Omnichannel Insights Organizations Need to Win in the Next Era of Agentic AI

SAP Unleashes Several "Ready-to-Use" AI Agents, Introduces a Custom Builder

The fresh tools will help the vendor "deliver the full promise of AI agents"

From Foundation Models to Agentic Mesh



The Agentic AI Tech Stack

Integration & Connectivity Layer
Protocol support
Service & agent registry
API management

Safety & Control Layer
Guardrails
Security & Compliance

Memory & Context Layer
Short-term memory
Long-term memory
Context management

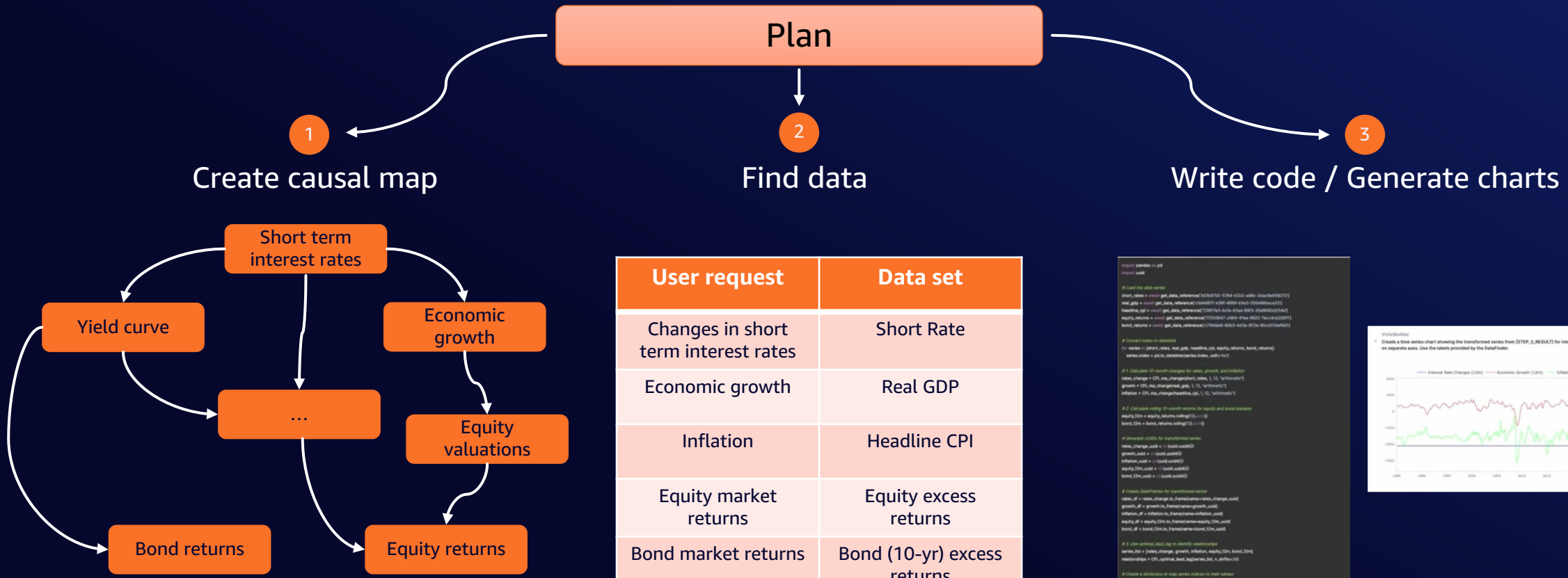
Development & Orchestration Layer
Agent Frameworks
Multi-agent orchestration
Evaluation Framework

Intelligence & Reasoning Layer
Large Language Models
Specialized task model

Infrastructure and Operations Layer
Model Serving, Vector Stores, Agent Hosting, Identity, Monitoring & Observability

Multi-agent investment analyst

How do changes in short-term interest rates influence growth and inflation? What are the subsequent effects on equity and bond market returns?



User request	Data set
Changes in short term interest rates	Short Rate
Economic growth	Real GDP
Inflation	Headline CPI
Equity market returns	Equity excess returns
Bond market returns	Bond (10-yr) excess returns



Financial market simulation



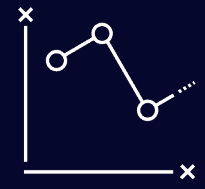
Market trend



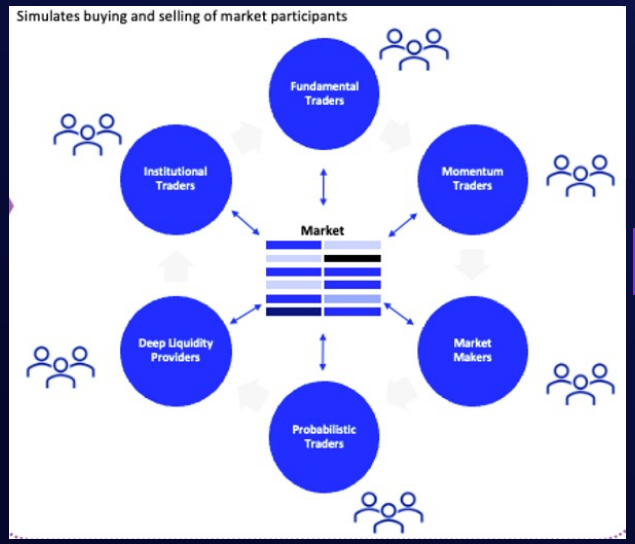
News & Social media



Economic policies



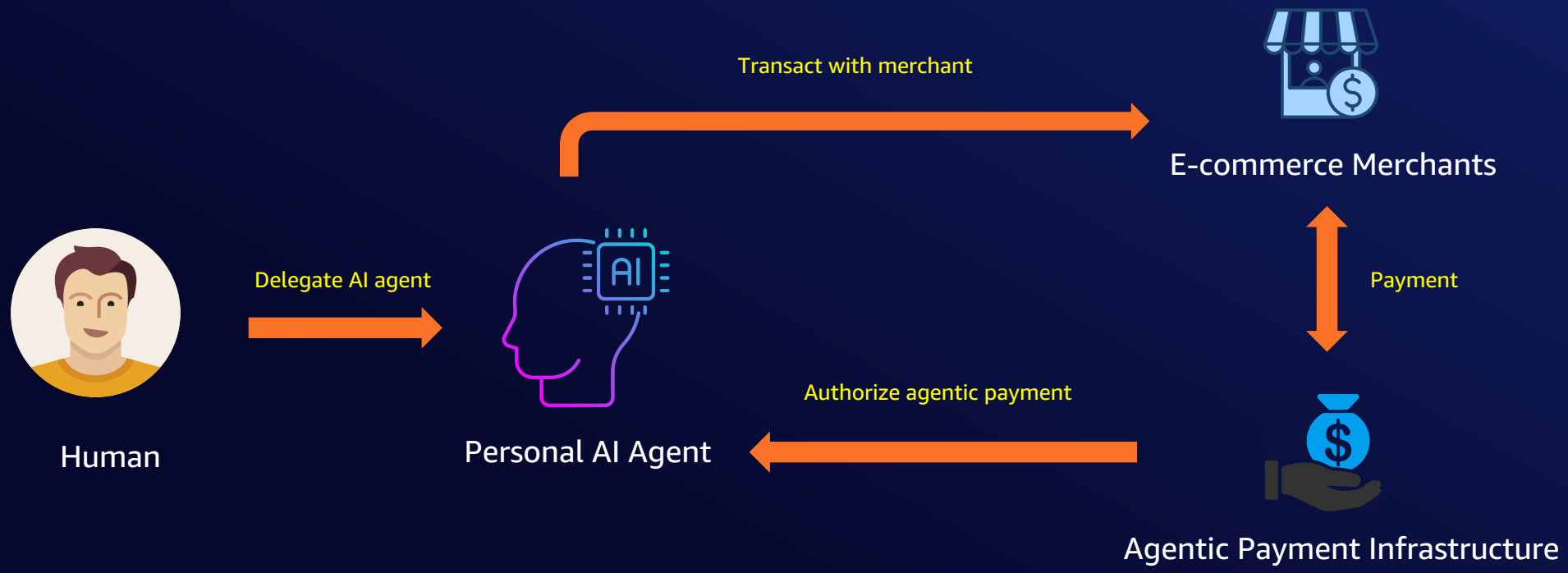
Extreme events



Benefits

1.Synthetic data generation for backtesting of extreme and plausible events
2.Market behaviors analysis under different risk scenarios
3.Controlled experimentations for different trading strategy

The Rise of Agentic Commerce



Agentic Mesh Ecosystem

The Marketplace helps users find agents that suit their needs, initiate and track requests, and provide feedback to Agents. Users initiate requests with agents and provide oversight to their actions through the Marketplace

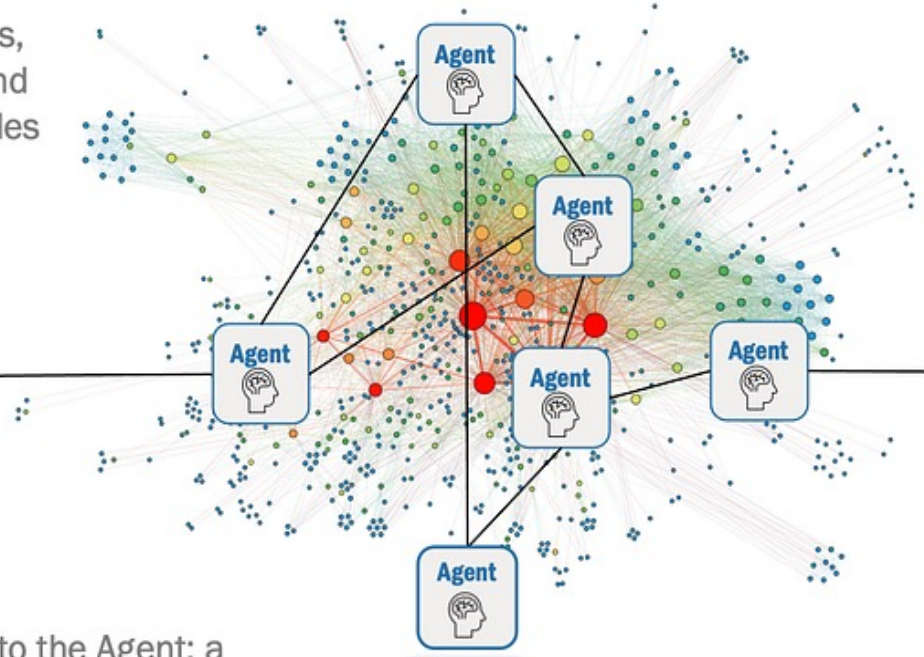
The Registry maintains Agent metadata (purpose, owner details, policies, security roles, capabilities, endpoint descriptions, and lifecycle states); It provides core discovery services.

An Ecosystem of Agents interact with each other to fulfill user requests

DNS maps human-readable names (example: agent.company.com) to Agent IP addresses/URLs. It allows agents can be located and connected to/from anywhere

Marketplace

Registry

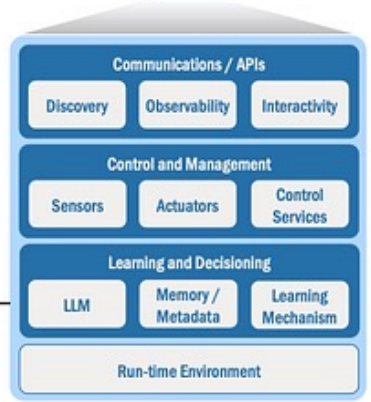


Internet DNS

Two types of language models are typically available to the Agent: a model (typically a Large Language Model) that is expert in creating step by step execution paths across many domains, and a local model that supports the Agents specialist capabilities

Orchestration LLM

Specialist LLM



An Agent Stack has the components required to autonomously fulfill tasks and engage with both other agents and users as needed.

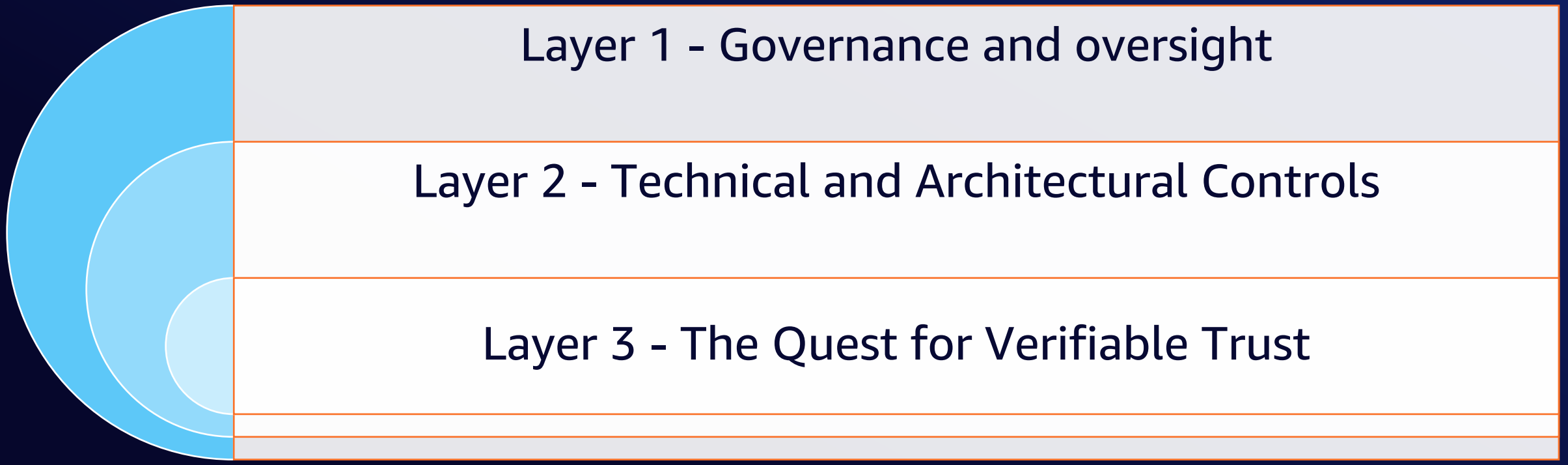
OWASP Agentic AI – Threats and Mitigations

1. MEMORY POISONING
2. TOOL MISUSE
3. PRIVILEGE COMPROMISE
4. RESOURCE OVERLOAD
5. CASCADING HALLUCINATION ATTACKS
6. INTENT BREAKING & GOAL MANIPULATION
7. MISALIGNED & DECEPTIVE BEHAVIORS
8. REPUDIATION & UNTRACEABILITY
9. IDENTITY SPOOFING & IMPERSONATION
10. OVERWHELMING HUMAN IN THE LOOP
11. UNEXPECTED RCE AND CODE ATTACKS
12. AGENT COMMUNICATION POISONING
13. ROGUE AGENTS IN MULTI-AGENT SYSTEMS
14. HUMAN ATTACKS ON MULTI-AGENT SYSTEMS
15. HUMAN MANIPULATION

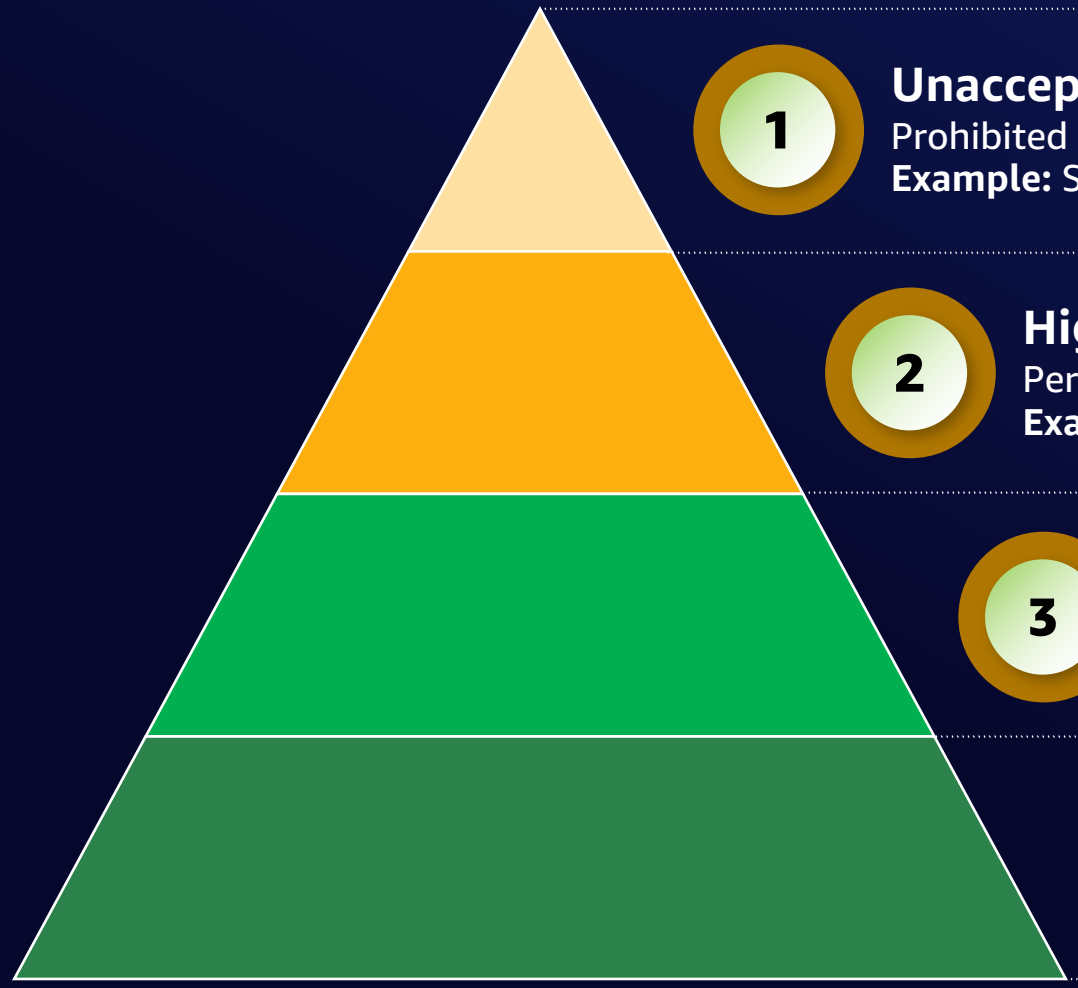
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Source: https://genai.owasp.org/resource/agentic-ai-threats-and-mitigations/

Multi-layers defense of Agentic AI security



EU AI Act



1

Unacceptable risk AI systems
Prohibited
Example: Social scoring

2

High risk AI systems
Permitted, subject to compliance with AI requirements ex-ante conformity assessment
Example: Recruitment

3

Limited risk AI systems
Permitted, but subject to information/transparency obligations
Example: Impersonation (bots)

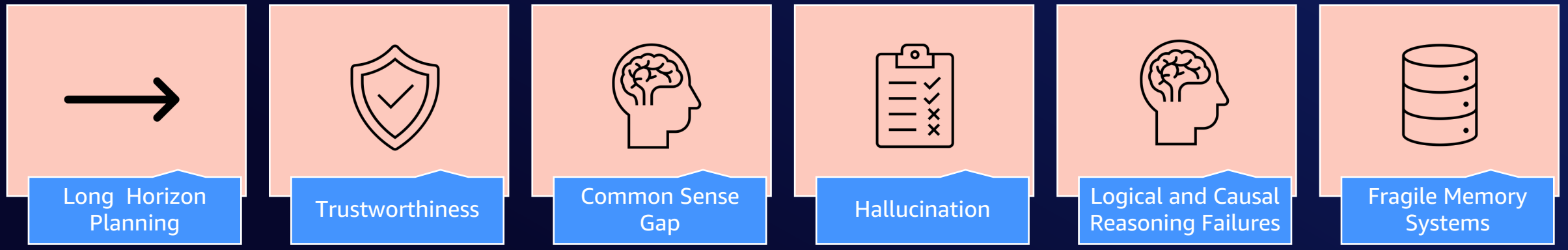
4

Minimal/no risk AI systems
Permitted without restrictions
Example: Predictive maintenance

Source: Wall Street Journal

AI Agent Challenges and Solution Paradigms

Challenges and Systemic Limitations in AI Agent Reasoning



Paradigms of AI reasoning in LLM based agents

- Foundational Prompting Strategies: The Emergence of Chain-of-Thought (CoT) Reasoning
- Action-integrated reasoning: The ReAct and ReWOO Frameworks
- Exploratory and deliberative reasoning: Tree of Thoughts (ToT) and Graph-of-Thoughts (GoT)
- Learning through trial and error : The Role of Reinforcement Learning
- Fusing logic and learning: The Advance of Neuro-Symbolic Architectures

Prepare for the agentic world

The Jobs That Are Changing

Customer service	Financial analysis	Project management	Software engineering
• From answering calls to designing agent experiences	• From creating reports to interpreting agent insights	• From tracking tasks to orchestrating AI workflows	• From writing coding to architecting and product design

90% of code will be created by AI by end of 2025

Why 25 Percent of Google's Code is Written by AI – And What it Means for Your Business

Stephen Gray • November 6, 2024 • Case Studies, Technology

A quarter of startups in YC's current cohort have codebases that are almost entirely AI-generated

Ivan Mehta – 2:10 AM PST · March 6, 2025

AI

Anthropic's CEO says that in 3 to 6 months, AI will be writing 90% of the code software developers were in charge of

Kwan Wei Kevin Tan Mar 13, 2025, 11:27 PM PT Share Save

Prepare for the agentic world

- **Go Beyond Prompt Engineering**: It's a skill, but it's not a career. Focus on the fundamentals that endure.
- **Think in Systems**: Learn how to build the "plumbing" around the LLM – the data pipelines, the APIs, the evaluation frameworks, the security controls. This is where the real engineering challenge lies.
- **Embrace Product Thinking**: Don't just build a cool demo. Start with a real user problem. Ask "Why?" before you ask "How?".
- **Develop "T-Shaped" Skills**: Deep expertise in one area (e.g., ML, backend systems, data science) combined with a broad understanding of the entire stack and the business context.

“The future is already here. It’s just not evenly distributed.”

Thank You!